

Problem-Solving Exercises

Coding Theory

SOLUTIONS

Question 1

1.a

$$\begin{aligned}\Pr\{y_j\} &= \Pr\{y_j \cap x_0\} + \Pr\{y_j \cap x_1\} = \Pr\{y_j|x_0\}\Pr\{x_0\} + \Pr\{y_j|x_1\}\Pr\{x_1\} \\ &= \frac{1}{2}(\Pr\{y_j|x_0\} + \Pr\{y_j|x_1\}).\end{aligned}$$

So, we have

$$\begin{aligned}\Pr\{y_0\} &= \frac{1}{2}(\Pr\{y_0|x_0\} + \Pr\{y_0|x_1\}) = \frac{1-p}{2}; \\ \Pr\{y_1\} &= \frac{1}{2}(\Pr\{y_1|x_0\} + \Pr\{y_1|x_1\}) = \frac{2p}{2} = p; \\ \Pr\{y_2\} &= \frac{1}{2}(\Pr\{y_2|x_0\} + \Pr\{y_2|x_1\}) = \frac{1-p}{2}.\end{aligned}$$

[5 marks]

1.b

We have

$$\begin{aligned}C &= \sum_{i=0}^1 \sum_{j=0}^2 \Pr\{x_i \cap y_j\} \cdot \log_2 \left(\frac{\Pr\{y_j|x_i\}}{\Pr\{y_j\}} \right) \\ &= \sum_{i=0}^1 \sum_{j=0}^2 \Pr\{y_j|x_i\}\Pr\{x_i\} \cdot \log_2 \left(\frac{\Pr\{y_j|x_i\}}{\Pr\{y_j\}} \right) = \frac{1}{2} \sum_{i=0}^1 \sum_{j=0}^2 \Pr\{y_j|x_i\} \cdot \log_2 \left(\frac{\Pr\{y_j|x_i\}}{\Pr\{y_j\}} \right),\end{aligned}$$

which can then be developed as

$$\begin{aligned}C &= \frac{1}{2} \left(\Pr\{y_0|x_0\} \cdot \log_2 \left(\frac{\Pr\{y_0|x_0\}}{\Pr\{y_0\}} \right) + \Pr\{y_0|x_1\} \cdot \log_2 \left(\frac{\Pr\{y_0|x_1\}}{\Pr\{y_0\}} \right) + \Pr\{y_1|x_0\} \cdot \log_2 \left(\frac{\Pr\{y_1|x_0\}}{\Pr\{y_1\}} \right) + \right. \\ &\quad \left. \Pr\{y_1|x_1\} \cdot \log_2 \left(\frac{\Pr\{y_1|x_1\}}{\Pr\{y_1\}} \right) + \Pr\{y_2|x_0\} \cdot \log_2 \left(\frac{\Pr\{y_2|x_0\}}{\Pr\{y_2\}} \right) + \Pr\{y_2|x_1\} \cdot \log_2 \left(\frac{\Pr\{y_2|x_1\}}{\Pr\{y_2\}} \right) \right).\end{aligned}$$

Finally, we obtain

$$\begin{aligned}C &= \frac{1}{2} \left((1-p) \cdot \log_2 \left(2 \frac{1-p}{1-p} \right) + 0 \cdot \log_2(0) + p \cdot \log_2 \left(\frac{p}{p} \right) + p \cdot \log_2 \left(\frac{p}{p} \right) + 0 \cdot \log_2(0) + \right. \\ &\quad \left. (1-p) \cdot \log_2 \left(2 \frac{1-p}{1-p} \right) \right) = \frac{1}{2} \left((1-p) \cdot \log_2(2) + 0 \cdot \log_2(0) + p \cdot \log_2(1) + \right. \\ &\quad \left. p \cdot \log_2 \left(\frac{p}{p} \right) + 0 \cdot \log_2(0) + (1-p) \cdot \log_2(2) \right) = 1-p.\end{aligned}$$

[10 marks]

Question 2

2.a

$$\begin{aligned}\Pr\{\text{more than 2 errors}\} &= 1 - \Pr\{\text{no error}\} - \Pr\{\text{one error}\} = 1 - \binom{l}{0}(1-p)^l - \binom{l}{1}p(1-p)^{l-1} \\ &= 1 - (1-p)^l - lp(1-p)^{l-1} = 1 - (1-p)^{l-1}(1-p+lp).\end{aligned}$$

Assuming $p \ll 1$, we obtain

$$\begin{aligned}\Pr\{\text{more than 2 errors}\} &= 1 - (1-p)^l - lp(1-p)^{l-1} \sim 1 - (1-lp) - lp(1-(l-1)p) = 1 - \\ &= 1 + lp - lp + l(l-1)p^2 = l(l-1)p^2.\end{aligned}$$

[10 marks]

2.b

For successful decoding based on the bit-flipping algorithm, we must minimise $\Pr\{\text{more than 2 errors}\} \sim l(l-1)p^2$, i.e., minimise the value of the parameter l . This simply means that each check node must be connected to the smallest possible number of bit nodes.

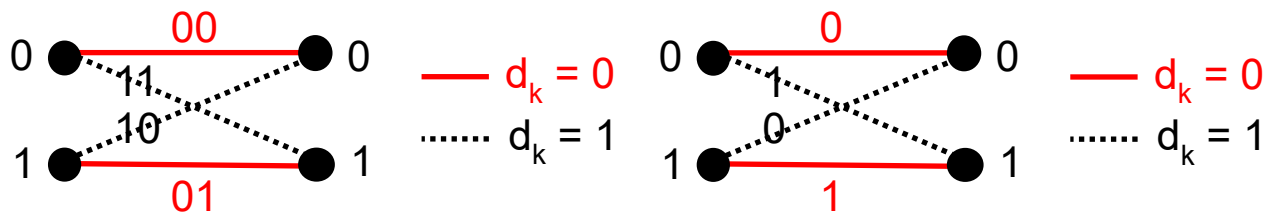
This implies that the number of 1s in each row of the parity-check matrix defining the linear block code must be as low as possible, even though there are many columns in that matrix. In other words, the parity-check matrix must contain very few 1s and many 0s. This is the definition of an LDPC code.

The bit-flipping algorithm can display good error-correcting capabilities when used in conjunction with an LDPC code.

[10 marks]

Question 3

3.a



[5 marks]

3.b Consider weight-2 messages at the input of the turbo encoder.

If the two 1s are separated by L 0s, the weight of the codeword generated by the first RSC encoder is equal to $d_L = 3 + L$.

If the two 1s are separated by L' 0s, the weight of the codeword generated by the second RSC encoder is equal to $d_{L'} = 1 + L'$.

The weight of the codeword generated by the turbo encoder is thus $d_{L,L'} = 4 + L + L'$.

Number of weight-2 messages separated by L 0s is given by $N_L \sim k$, when $k \gg L$.

Number of weight-2 messages separated by L' 0s is given by $N_{L'} \sim k$, when $k \gg L'$.

Probability to have a weight-2 message separated by L' 0s after random interleaving is $\Pr\{L'\} \sim \frac{k}{\binom{k}{2}} \sim \frac{k}{\frac{k^2}{2}} = \frac{2}{k}$, when $k \gg L'$.

Number of weight-2 messages separated by L 0s that become a weight-2 message separated by L' 0s after random interleaving is $N_{L,L'} = N_L \Pr\{L'\} \sim \frac{2k}{k} = 2$, when $k \gg L, L'$.

Weight-4 codeword obtained with $(L, L') = (0,0) \rightarrow e(d=4) = \frac{w_{d=4}}{2k} \sim \frac{4}{2k} = \frac{2}{k}$.

Weight-5 codeword obtained with $(L, L') = (1,0), (0,1) \rightarrow e(d=5) = \frac{w_{d=5}}{2k} \sim \frac{8}{2k} = \frac{4}{k}$.

Weight-6 codeword obtained with $(L, L') = (1,1), (2,0), (0,2) \rightarrow e(d=6) = \frac{w_{d=6}}{2k} \sim \frac{12}{2k} = \frac{6}{k}$.

Finally, the first three terms in the union bound are given by

$$P_{eb} \sim \frac{2}{k} \operatorname{erfc} \left(\sqrt{\frac{4}{3} \frac{E_b}{N_0}} \right) + \frac{4}{k} \operatorname{erfc} \left(\sqrt{\frac{5}{3} \frac{E_b}{N_0}} \right) + \frac{6}{k} \operatorname{erfc} \left(\sqrt{\frac{6}{3} \frac{E_b}{N_0}} \right).$$

This is not a very good turbo encoder as its minimal distance is quite low ($d_{min} = 4$ only), especially given its low coding rate, but it still works well in the sense that as it does achieve interleaver gain (thus resulting in very low error coefficient values).

[10 marks]

3.c

[5 marks]

Question 4

Coding rate R_c	Minimum SNR per Shannon, $\frac{E_b}{N_0}$, in dB, required to achieve error-free communications
0.5	$\sim -2.8 - 10\log_{10}(0.5) \sim 0.2$
0.6	$\sim -1.6 - 10\log_{10}(0.6) \sim 0.6$
0.7	$\sim -0.4 - 10\log_{10}(0.7) \sim 1.1 - 1.2$
0.8	$\sim 1.0 - 10\log_{10}(0.8) \sim 2.0$
0.9	$\sim 2.7 - 10\log_{10}(0.9) \sim 3.1 - 3.2$

[10 marks]

Question 5

5.a

Since both codes are linear, we can search for the codewords with the smallest Hamming weight to determine d_{min} .

When each row of a non-zero message is encoded by C_1 , the resulting codeword contains at least one row with a Hamming weight equal to $d_{min,1}$. Having one row with $d_{min,1}$ ones and all other rows filled with zeros is actually a worst-case scenario that can occur only when all the ones in the message are inside the same row. This worst-case configuration results in an array in which there are $d_{min,1}$ columns containing a single one.

When each column is then encoded by C_2 , we obtain a codeword in which there are at least $d_{min,1}$ columns having a minimum Hamming weight of $d_{min,2}$. The minimum Hamming weight of the codeword is thus $d_{min} = d_{min,1}d_{min,2}$.

[10 marks]

5.b

Think about a decoding strategy where both constituent codes are decoded sequentially, and the extrinsic information generated by the second decoder is then sent back to the input of the first decoder. This is an iterative decoding process similar to that used in turbo decoding or the sum-product algorithm employed by LDPC decoders.

[10 marks]

Question 6

The method consists of setting one bit node at a time to 1, and then scanning all possibilities in the Tanner graph regarding the values that must be taken by all other bit nodes to satisfy all check nodes. This search is obviously very time-consuming when performed manually in the context of large Tanner graphs, but it can be automated easily. The minimum Hamming distance is associated with the acceptable bit node configuration(s) having the smallest Hamming weight.

Figure 4: We find at least one possible configuration in the Tanner graph with a Hamming weight of 2. Therefore, we have $d_{min} = 2$.

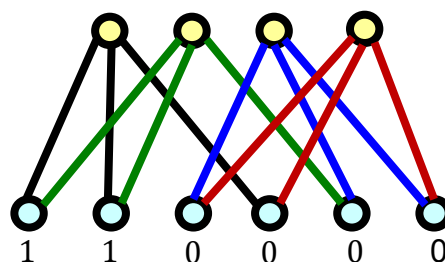


Figure 5: There can be only one codeword different from the all-zero codeword. Its Hamming weight is 5. Therefore, we have $d_{min} = 5$.

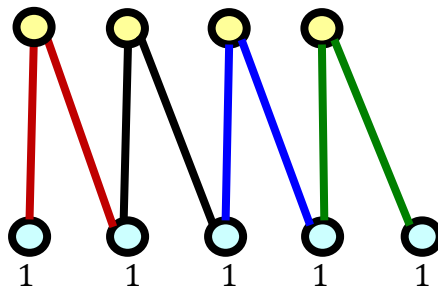


Figure 6: We find many possible configurations in the Tanner graph with a Hamming weight of 3, but there is no configuration with a Hamming weight of 2. Therefore, we have $d_{min} = 3$. An example is shown below.

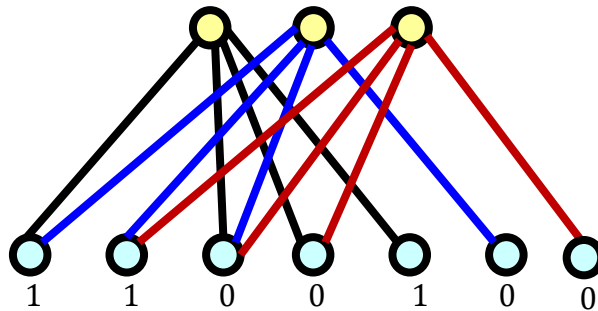
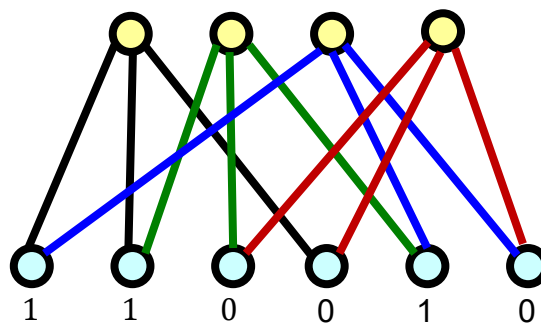


Figure 7: We find many possible configurations in the Tanner graph with a Hamming weight of 3, but there is no configuration with a Hamming weight of 2. Therefore, we have $d_{min} = 3$. An example is shown below.



[15 marks]

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